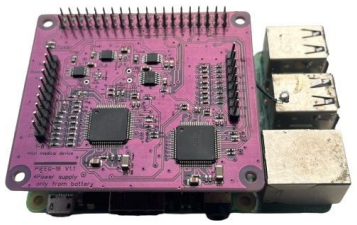
CONNECTOR PINOUT & ELECTRODE GUIDE · SKU CQA240229G16

Electrodes, Headers & GPIO Pins

How to connect 16 electrodes, the reference and the bias to PiEEG-16, a suggested whole-head 10-20 montage, and the seven Raspberry Pi GPIO pins the two converters use.



16	2 × 8	REF	BIAS	7
Electrode channels	One group per ADS1299	Ear clip · common 0 µV	Ear clip · noise rejection	GPIO signal pins used

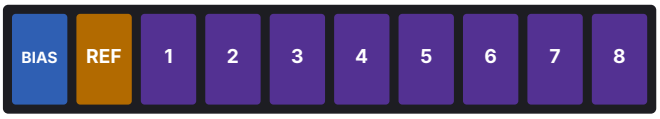
01 ELECTRODE INPUTS

Input	Count	Read by	Purpose
Channels 1–8	8	ADS1299 #1	Left header; voltage between each electrode and the reference
Channels 9–16	8	ADS1299 #2	Right header; same reference
Reference	1	Both	Ear clip on one earlobe; common 0 µV point for all 16 channels
Bias	1	Both	Ear clip on the other earlobe; cancels common-mode noise such as 50/60 Hz hum

02 ELECTRODE HEADERS

Left header · 10 pins

BIAS + REF ear clips, channels 1–8 (ADS1299 #1)



Right header · 8 pins

channels 9–16 (ADS1299 #2)



Group view. 2.54 mm pitch. Follow the channel numbers printed next to each pin on the board for the exact pin order.

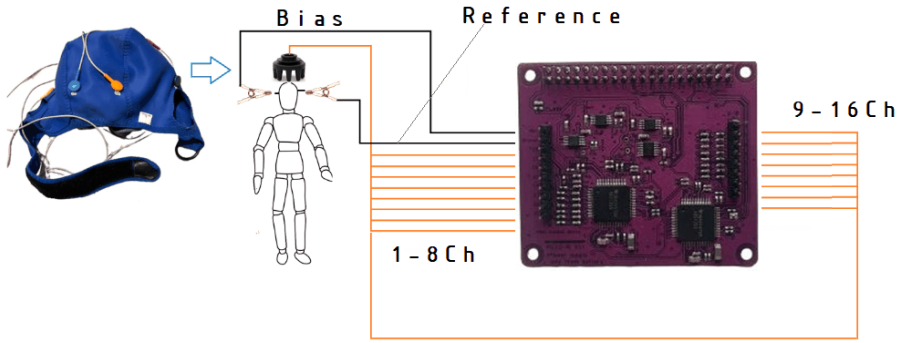


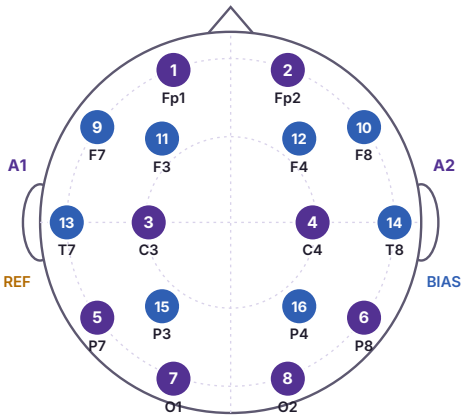
Figure 1 — Wiring: channels 1–8, reference and bias on the left header; channels 9–16 on the right header.

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NOTE Reference and bias clips matter most.

A loose ear clip makes all 16 channels noisy at once. If only channels 1–8 or only 9–16 look wrong, check that group's electrodes first.

03 SUGGESTED MONTAGE (10-20 SYSTEM)

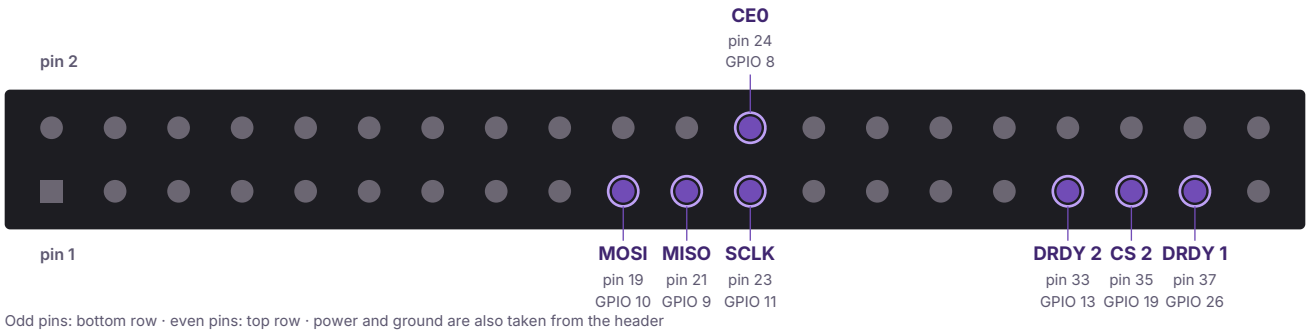


Ch 1–8 · ADS1299 #1 Ch 9–16 · ADS1299 #2

Ch	Site	Area	Ch	Site	Area
1	Fp1	Front left	9	F7	Front left, side
2	Fp2	Front right	10	F8	Front right, side
3	C3	Centre left	11	F3	Front left
4	C4	Centre right	12	F4	Front right
5	P7	Back left, side	13	T7	Left temple
6	P8	Back right, side	14	T8	Right temple
7	O1	Back left	15	P3	Back left, top
8	O2	Back right	16	P4	Back right, top

Channels 1–8 match the PiEEG-8 default, so recordings from both boards compare directly. Any other positions work — note which channel went where.

04 RASPBERRY PI GPIO PINS



Odd pins: bottom row · even pins: top row · power and ground are also taken from the header

Header pin	GPIO	Signal	Chip
19	GPIO 10	SPI0 MOSI	Both
21	GPIO 9	SPI0 MISO (DOUT)	Both
23	GPIO 11	SPI0 SCLK	Both
24	GPIO 8	SPI0 CE0 (chip select)	#1 (ch 1–8)
35	GPIO 19	Chip select, driven by software	#2 (ch 9–16)
37	GPIO 26	DRDY 1, active low	#1
33	GPIO 13	DRDY 2, active low	#2

The shield also takes power and ground from the header. Other GPIO pins are free for your own triggers or buttons, if you can reach them with the shield mounted.

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05 FITTING THE CAP

Step	Action
1	Measure the head and pick the right cap size; find **Cz** and align the cap.
2	**Dry electrodes:** part the hair under each electrode. **Wet:** work gel through the hair with the blunt syringe.
3	Attach the ear clips to clean earlobes: REF on one, BIAS on the other.
4	Bundle the 16 leads, keep them short and still, and ask the person to relax the face and jaw.

06 RELATED DOCUMENTS

PE16-DS-001 Product Datasheet	**PE16-MD-002** Mechanical & Dimensional Drawing	**PE16-PCB-003** PCB & Manufacturing Specification
PE16-QSG-005 Quick Start & Safety Guide	**PE16-SW-006** Data Interface & Software Guide	**PE16-ENC-007** 3D-Printed Enclosure Guide